

Week 5 — Interaction & Effect Modification

SPPH 604 · Application of Advanced Epidemiological Methods

Running example: Kueh et al. (2026), MASLD / NHANES

Interaction & Effect Modification

Week 5 · Tuesday lecture — the concept lecture before Thursday's **L4** lab

Three ideas students collapse into one:

- ▶ **Confounding** — a nuisance to remove
- ▶ **Effect modification** — a finding to report
- ▶ **Statistical interaction** — a model term on a scale
MASLD: does the Group IV survival penalty differ by sex
— or does it just look different?

Learning objectives

- ▶ Distinguish **confounding**, **effect modification (EMM)**, and **statistical interaction**
- ▶ Explain why EMM is **scale-dependent** — additive vs multiplicative answer different questions
- ▶ Read the MASLD **sex** example correctly: a crude gap is **not an interaction test**, and a null test is **not homogeneity**
- ▶ Pick the **right tool for time-to-event** — a Cox product term + absolute-risk contrasts
- ▶ Reject the logistic **RERI** for a survival outcome, and say why
- ▶ Name the term's dominant threat you will formalize at **M1**: **FLI circularity**

Continuity — where this sits

- ▶ **Last week (Wk4 / L3):** we adjusted confounders and built **Table 1** (P2) — the Group IV HR shrank from **15** → **2.9**
- ▶ Confounders **distort** one honest effect; we removed them
- ▶ **Today:** a *different* question — does the effect **differ across subgroups**?
- ▶ **Thursday (L4):** product terms + stratum-specific estimates, which each of you runs as **P3**
***MASLD:** last week adjustment shrank the IV effect; today we ask whether that effect is the same for men and women.*

Where we are in the project

- ▶ Course arc: **Replicate** → **Interrogate** → **Improve** → **Defend** — we are mid-**Interrogate**
- ▶ **P1** eligibility · **P2** Table 1 · today feeds **P3** **effect modification**
- ▶ The finding to keep in view: **Group IV** (non-obese, high central adiposity) has the **worst all-cause survival**
- ▶ Four phenotypes stay on screen all day as our running example **MASLD**: *keep the four BMI × WHtR groups as our test bed — ours are I 391, II 4,585, III 787, IV 608; the paper's are 386 / 4,541 / 769 / 604.*

Today's roadmap

- ▶ **Three concepts → two scales → one worked MASLD example → the right time-to-event tool**
- ▶ Flag the trap up front: a big crude men-vs-women gap is **not** proof of modification
- ▶ We will vote on the sex example, then dismantle it together
- ▶ Ends with honest subgroup reporting + a preview of the term's dominant threat
***MASLD:** by the slide after the break we test the **men-IV 28 vs women-IV 4** gap properly — and find that the data cannot say what produced it.*

Three things people call “interaction”

- ▶ **Confounding** — a nuisance *common cause*; we adjust it **away** to recover one honest effect
- ▶ **Effect modification (EMM)** — the effect **genuinely differs** across a third variable; a *finding*, not a nuisance
- ▶ **Statistical interaction** — a **model term** (product coefficient); a device that *may or may not* map to real EMM
- ▶ Same word, three jobs — conflating them is the root error **MASLD**: *for the Group IV effect, sex could plausibly be any of the three — today we test which.*

Confounding vs effect modification

- ▶ Two different questions: “*is my single estimate biased?*” vs “*should there even **be** a single estimate?*”
- ▶ You **remove** confounding; you **report** effect modification (stratum-specific estimates)
- ▶ Diagnostic contrast: adjusting a confounder **moves the pooled estimate**; a true modifier makes a **pooled estimate misleading**
- ▶ A confounder is a problem to fix; a modifier is a result to describe
***MASLD:** the paper adjusts sex **as a confounder** — we ask instead whether sex **modifies** the Group IV effect.*

Statistical interaction = a product term

- ▶ Model form: `outcome ~ exposure + modifier + exposure:modifier`
- ▶ The **product coefficient** *is* the interaction — but only **on that model's scale**
- ▶ Meaning is inherited from the **link**: log-hazard, log-odds, or risk
- ▶ A **null product term does not mean equal absolute effects**
***MASLD:** in Cox, the term is `Group:Sex` on the **log-hazard (multiplicative) scale**.*

Additive vs multiplicative scales

- ▶ **Multiplicative:** do the **ratios** differ across strata? (ratio of HRs / RRs)
- ▶ **Additive:** do the **risk differences** differ? — governs *how many events you prevent*
- ▶ Public-health / intervention planning lives on the **additive** scale
- ▶ Two men and women can share an HR yet differ hugely in **absolute** excess deaths
MASLD: same *HR* + different **baseline** mortality → very different absolute excess deaths by sex.

Same data, two answers

- ▶ Toy $2 \times 2 \times 2$: a setting with **no multiplicative** interaction but **clear additive** interaction (and vice versa)
- ▶ Takeaway: “*is there effect modification?*” is **unanswerable without naming the scale**
- ▶ Reporting rule: state the scale, give stratum-specific estimates, don't hang it all on one p-value
- ▶ The scale you pick is a *choice*, and it changes the verdict
- ▶ Multiplicative-null additive-null

MASLD: differing **baseline rates** are one candidate explanation for the sex “signal” — hold that thought, and note we will not be able to confirm it.

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- ▶ So a single “no interaction” claim is incomplete
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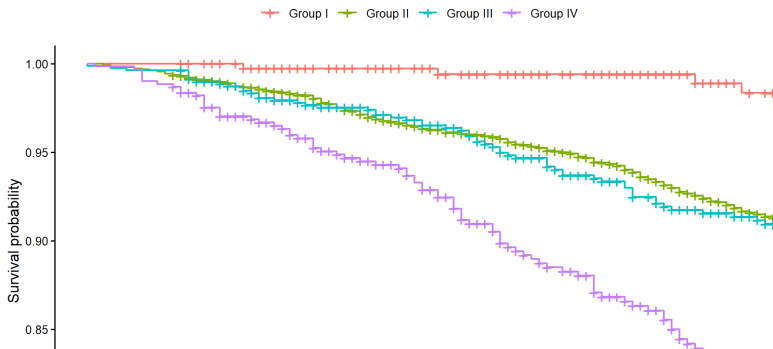
BREAK (10 min)

- ▶ Stretch — back in **10 minutes**
- ▶ On return: the **MASLD sex example** — men-IV 28 vs women-IV 4
- ▶ Come back ready to **vote**: modification, confounding, or noise?

The MASLD sex question

- ▶ A reader's claim: *"the Group IV hazard is far worse in **men**."*
- ▶ Formal question: does **sex modify** the Group IV-vs-reference **hazard ratio**?
- ▶ Design: **time-to-event**, **586** all-cause deaths in our reproduction (paper: **585**), median follow-up **6.7 y**
- ▶ We test sex specifically for the **Group IV** contrast — the paper's headline

All-Cause Mortality



Crude stratified hazard ratios

- ▶ Sex-stratified **crude** Group IV HRs: **men 28** vs **women 4**
— visually dramatic
- ▶ But crude stratum HRs mix in **confounding** *and* **differing baseline hazards** *and* **small male cell counts**
- ▶ Ask the room — is this **modification, confounding, or noise?** (*hold the vote*)
- ▶ Resist the headline reflex
MASLD: *these are the exact numbers a careless subgroup analysis would put in the abstract.*

The formal interaction test

- ▶ Fit the Group:Sex product term in the **crude** Cox model, then the **adjusted** model
- ▶ Match the test to the estimand. A group*sex term across four groups is a **multi-df** test of whether *any* contrast varies by sex. For the locked IV-vs-III question the 1-df test gives a ratio of hazard ratios of **1.00 (0.42–2.35)**, **p = 0.99** — no evidence of interaction, but an interval wide enough to be compatible with substantial heterogeneity. A null p-value does not prove homogeneity.
- ▶ Lesson: crude stratum contrasts are **not** an interaction test; the test must live **in a model, adjusted**
- ▶ A borderline crude p was never evidence of modification
MASLD: on the locked contrast there is no evidence of interaction at the available precision — which is not a finding that the men-28 / women-4 gap was illusory.

A crude stratum difference is not evidence of modification

- ▶ **Do not explain the crude gap.** A p-value that moves when covariates enter tells you the **estimate** changed — not **what generated** the difference
- ▶ **Withdrawn:** the earlier teaching that this was baseline-risk confounding rather than true modification, and the instruction to report a **common multiplicative effect** across sex
- ▶ Differing baseline mortality is **one candidate** explanation among several — this study cannot distinguish them
- ▶ The honest statement: **1.00 (0.42–2.35)** — no evidence of interaction **at the available precision**, on an interval compatible with **less than half** the effect in women, or **more than twice** it
MASLD: a null with its uncertainty attached, not a demonstration that sex is irrelevant. M1 re-runs it design-aware on the same rows and the picture moves.

The right tool for time-to-event

- ▶ **Multiplicative test:** a **product term** in the **Cox model** (Group:Sex)
- ▶ **Additive story:** **absolute-risk / survival contrasts** — predicted risk differences at fixed times
- ▶ **Do not** dichotomize survival and run a logistic **RERI** — it discards follow-up time and censoring
- ▶ If additive interaction is the target, get it from model-based **absolute risks**, not a repurposed odds ratio
***MASLD:** pair the Group IV **HR** with predicted **6-year risk differences by sex** to serve public-health readers.*

Honest subgroup reporting

- ▶ **Power:** interaction tests are underpowered — a null is **weak** evidence of *no* modification (tiny male IV cell)
- ▶ **Multiplicity:** **pre-specify** the modifier; don't fish across every covariate
- ▶ **Report:** stratum estimates + interaction p + the scale — not a single “significant subgroup”
- ▶ One honest pre-specified test beats twenty exploratory ones
***MASLD:** the sex example is the cautionary template for what you must resist in **P3**.*

Preview: the term's dominant threat

- ▶ Seeds the idea you will formally name at **M1: FLI circularity**
- ▶ The phenotype groups use **BMI + waist**; the FLI eligibility filter *also* uses **BMI + waist** — exposure and cohort selection are **entangled**
- ▶ Why it matters here: an EMM claim is only as trustworthy as the **exposure definition**
- ▶ Tease the check: a **TyG-based** sensitivity exposure that removes **BMI and waist** from the steatosis surrogate — MASLD's cardiometabolic criteria can still involve them
***MASLD:** flag it now so your **P3** subgroup work is read against the looming threat, not in isolation.*

Bridge to today's lab (L4) & P3

- ▶ **In L4 (Thu, EpiMethods OER)**, on the fixed NHANES teaching dataset you will:
 - ▶ (a) fit a Cox model with an **exposure × modifier product term**
 - ▶ (b) extract **stratum-specific** hazard ratios
 - ▶ (c) read the **interaction p-value** on the multiplicative scale
 - ▶ (d) compute **absolute-risk contrasts** at a fixed time — deliberately **not** a logistic RERI
- ▶ **Deliverable P3:** one **pre-specified** effect-modification test on **your own paper's exposure** — product-term p (crude vs adjusted), stratum estimates, the scale, and a one-line verdict stated as **evidence, not mechanism**: whether the data show evidence of modification at the precision you have, and what the interval leaves open
- ▶ **P3 is optional and ungraded** — no separate weight, no separate deadline, nothing to submit; it applies L4's method to your own paper, and **M1 is assembled from these five increments**, the only place it is assessed
MASLD: your P3 should read like our sex example done honestly — report the interval, and resist headlining a